

Poisson Layers at the Finite-Core Boundary of Exponential-Feedback Transseries

An explicit missing factor, a logarithmically shifted critical window,
and joint fluctuations of the rare layer and the giant action

Research article prepared for Vladimir Reshetnikov
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Abstract

Consider the formal equation

$$\widehat{U}(q) = \sum_{j \geq 1} q^j \exp(a j^p \widehat{U}(q)), \quad a > 0, \quad p > 1.$$

A recent ProveIt manuscript proves that a core containing actions $1, \dots, M$, together with exactly one larger action, captures the full n th coefficient precisely when $M(p-1) > 1$. It leaves the distribution of the omitted background actions and finite transition corrections open. We prove a uniform marked-coefficient theorem throughout the larger range $(M+1)(p-1) > 1$. If $w_{n,M}$ is the exact one-tail coefficient and $u_n(s)$ marks each occurrence of action $M+1$ by s , then

$$u_n(s) = w_{n,M} \exp(s \nu_{n,M})(1 + o(1)), \quad \nu_{n,M} = \frac{n r_n}{1 + r_n} e^{-M p r_n},$$

uniformly for real s in compact subsets of $[0, \infty)$. Here $r_n(1 + r_n)^{p-1} e^{p r_n} = a n^{p-1}$. The result is uniform on compact parameter sets with a strict margin, and survives bounded nonnegative changes to every action inside the core. Thus one fewer action in the implicit core can be compensated by a single explicit exponential factor.

At the boundary $p = 1 + 1/M$, the missing factor is exactly $\exp(r_n^{M+1}/a^M)$ up to relative error tending to zero, not a constant. A finite Poisson limit instead occurs when $p = 1 + 1/M + ((M+1) \log \log n + \theta)/(M \log n)$; its parameter is $e^{-\theta}/(a^M (M+1)^{M+1})$. We prove total-variation convergence for bounded intensities, a central limit theorem and a large-deviation principle for diverging intensities, relative asymptotics for fixed rare counts, and independence from the Gaussian fluctuation of the giant action. The proof separates the existing finite-core method from the new uniform saddle perturbation. Exact coefficient identities, numerical diagnostics, and further research questions complete the article.

Editorial note (ProveIt, 2026-09-29). Scope: at the boundary itself, $p = 1 + 1/M$, the missing factor diverges (Theorem 6.1), so the finite-core ratio tends to zero there, as `Finite_Core_Universality_Exponential_Feedback/` states (its Theorem `thm:cutoff`). The finite Poisson limit of Theorem 7.1 occurs only in a moving window slightly *above* the boundary, outside that package's fixed-parameter theorem; it interpolates between its limits 0 and 1 and does not contradict it. This article sharpens that package: $\log \nu_{n,M}$ in (7.4) equals, term by term, its one-occurrence scale (its equation `eq:cutoff-scale`), whose lower bound is the first-order term of e^ν here. The total-variation Poisson law at diverging intensity, left open here (Section 11), is proved for $\lambda_j = j^2$ ($M = 1, p = 2, a = 1$) by `Microscopic_Condensation_Exponential_Feedback/`; both packages are in `Analysis/Transseries/docs/series-and-transseries/`.

Status and scope. This is an AI-assisted research draft with conventional mathematical proofs, not a Lean-verified development. The results are proposed additions to the inspected repository work. Global originality and publication priority have not been established. No general resurgence theorem or signed inverse conjecture is claimed solved.

Contents

1	The repository question and the contribution	3
2	Model, coefficient ensemble, and main theorem	4
3	Exact formal algebra and probability	5
4	Compact-uniform finite-core reduction	6
5	Two saddles and their uniform perturbation	8
6	Every cutoff boundary has an explicit missing factor	11
7	The logarithmically shifted Poisson window	12
8	Poisson, Gaussian, and rare-event consequences	13
9	The rare layer decouples from the giant action	14
10	Computation and checks	15
11	Further research questions	17
12	Conclusion	18
A	Proof-dependency and scope audit	19
B	Notation and an implementation checklist	20

1 The repository question and the contribution

1.1 A specific gap in the existing finite-core theory

The ProveIt repository now keeps its generic transseries work under `Analysis/Transseries/`. Its documentation distinguishes the canonical transseries volume, focused Lean modules, and recent unmerged research manuscripts [1]. The present paper continues *Finite-Core Universality and Sharp Large Order for Countable Exponential-Feedback Transseries* [2], rather than attempting to replace the entire transseries program by a new survey.

The inspected source is pinned to commit

0c973d8f5e7ef4550f4df1bf486d890037fd9f14.

Its file is `Finite_Core_Universality_Exponential_Feedback/article.tex` inside the transseries documentation directory, with blob `c2b6faf623003111e9d9e6bbb162a67429f3ce42`. Its research subsections “Regularly varying feedback and critical logarithmic boundaries” and “Joint laws for the finite background cloud” explicitly ask for finite nonzero corrections at a critical cutoff and for Poisson/Gaussian laws of the small actions. Its necessity argument treats only one additional occurrence of the first omitted action. A one-occurrence lower bound proves failure of a cutoff, but does not determine the entire missing factor.

We sum that layer to relative precision. The result answers the first-omitted- action part of the cloud question and supplies a moving-power critical window. It does *not* settle the proposed generalization to arbitrary regularly varying slopes. It also does not settle the signed quadratic inverse conjecture in the same source. These boundaries are important: positivity is used in the proof, and a fixed-parameter theorem cannot be substituted into a moving-parameter regime without establishing uniformity.

1.2 What is inherited and what is added

The positive Lagrange coefficient formula, the giant-action localization, the strict finite-core cutoff, and the two-stage saddle method originate in the repository manuscript [2]. We re-establish the forms needed here, with compact-uniform estimates. They are supporting arguments, not independent novelty claims.

The additions are the uniform marked ratio, its reduction to a scalar core-independent intensity, the explicit correction at every reciprocal cutoff boundary, the shifted Poisson window with its constant, and the joint rare-layer/giant-action limits. In particular, our coefficient equivalent is stronger than a logarithmic equivalent: an error $o(1)$ occurs *after* taking the logarithm of the ratio, not merely after dividing that logarithm by n or by the diverging intensity.

1.3 Relation to established mathematics

Lagrange inversion itself is classical; see Gessel [3]. Formal transseries provide the algebraic context, including the distinction between formal operations and analytic realization [4]. In the present model, the substitution $q = e^{-x}$ produces a one-generator exponential transseries. Its support is simple; the difficult phenomenon is the growth and organization of its coefficients.

Poisson layers, condensation, and Gaussian clouds are not new phenomena in probability. Janson, Jonsson, and Stefánsson prove Poisson boundary laws and joint Gaussian/Poisson limits for superexponentially weighted random trees [5]. Their weights are products of degree weights. Ours contain a power of the *sum* of action slopes and have a giant of order $n/\log n$. The logarithmic displacement of the cutoff and the scalar intensity below are features of this feedback model, not consequences of identifying the two probability ensembles.

There is also an established notion of mod-Poisson convergence [6]. We prove a positive-real-marker exponential approximation. We do not silently promote it to a relative approximation on an arbitrary complex marker domain when the intensity diverges. The probabilistic consequences claimed below are proved from the estimates actually available.

2 Model, coefficient ensemble, and main theorem

2.1 A finite core and a power tail

Fix an integer $M \geq 1$ and put $d = M + 1$. To display the universality precisely, allow a finite nonnegative core

$$\Phi_M(q, u) = \sum_{j=1}^M b_j q^j e^{\ell_j u}, \quad b_1 = 1, \quad b_j, \ell_j \geq 0. \quad (2.1)$$

For a real marker $s \geq 0$, let

$$\widehat{U}_s(q) = \Phi_M(q, \widehat{U}_s(q)) + s q^d e^{ad^p \widehat{U}_s(q)} + \sum_{j>d} q^j e^{aj^p \widehat{U}_s(q)}, \quad u_n(s) = [q^n] \widehat{U}_s(q). \quad (2.2)$$

Every infinite identity involving $\widehat{U}_s$ is formal. The pure model is $b_j = 1$, $\ell_j = aj^p$, with $s = 1$. No evaluation of the divergent infinite kernel at positive numerical arguments is used.

The parameters may vary with n . Uniformity will always mean

$$\begin{aligned} a &\in [a_-, a_+], & p &\in [p_-, p_+], & 0 < a_- \leq a_+ < \infty, \\ 1 < p_- \leq p_+ < \infty, & d(p_- - 1) > 1, \\ 0 \leq b_j, \ell_j \leq C_0 \quad (j \leq M), & b_1 = 1, & 0 \leq s \leq S < \infty. \end{aligned} \quad (2.3)$$

Here M, C_0, S and the compact intervals are fixed. This is a strict-margin condition; no assertion is made at $d(p_- - 1) = 1$.

2.2 The exact one-tail series

Let $G = G_M$ and $R = R_M$ be the small analytic branches

$$G(q) = \Phi_M(q, G(q)), \quad R(q) = \frac{1}{1 - \partial_u \Phi_M(q, G(q))}. \quad (2.4)$$

They satisfy $G(q) = q + O(q^2)$ and $R(q) = 1 + O(q)$, with nonnegative Taylor coefficients. Define

$$W_M(q) = R(q) \sum_{j>M} q^j e^{aj^p G(q)}, \quad w_{n,M} = [q^n] W_M(q). \quad (2.5)$$

The large-action sum in (2.5) is formal. The coefficient $w_{n,M}$ counts exactly one primitive action above M , not simply the coefficient of the finite truncation G_M .

2.3 The scalar intensity

For $n > 0$, let $r = r_n(a, p) > 0$ be the unique solution of

$$r(1 + r)^{p-1} e^{pr} = an^{p-1}. \quad (2.6)$$

Existence and uniqueness follow because the left side increases strictly from zero to infinity. Set

$$j_0 = \frac{n}{1+r}, \quad k_0 = \frac{nr}{1+r}, \quad t_0 = e^{-pr}, \quad \boxed{\nu_{n,M} = k_0 t_0^M = \frac{nr}{1+r} e^{-Mpr}}. \quad (2.7)$$

The intensity depends only on a, p, n, M , not on the finite core coefficients. The root equation is often numerically better evaluated after taking logs.

Theorem 2.1 (Uniform Poisson-layer coefficient theorem). *Under (2.3),*

$$\sup \left| \frac{u_n(s)}{w_{n,M} e^{s\nu_{n,M}}} - 1 \right| \rightarrow 0. \quad (2.8)$$

The supremum is over all parameters in that domain. In particular,

$$u_n(1) = w_{n,M} e^{\nu_{n,M}} (1 + o(1)), \quad u_n(0) = w_{n,M} (1 + o(1)). \quad (2.9)$$

These are multiplicative equivalents, including when $\nu_{n,M} \rightarrow \infty$.

Remark 2.2 (One fewer implicit core action). The earlier uncorrected criterion is $M(p-1) > 1$. The corrected formula requires only $(M+1)(p-1) > 1$. For $1 < p \leq 2$, the smallest sufficient uncorrected core has $\lfloor 1/(p-1) \rfloor + 1$ actions. The present formula uses $\lfloor 1/(p-1) \rfloor$ core actions plus the scalar factor $e^{\nu_{n,M}}$. It does not remove the need to evaluate the one-tail coefficient or its saddle equivalent.

2.4 A directly computable coefficient equivalent

The one-tail coefficient has a finite-core saddle formula. Let (z, τ) be the unique saddle in its small-radius asymptotic region:

$$n = z + az^p \tau G'(\tau), \quad -\log \tau = apz^{p-1} G(\tau). \quad (2.10)$$

Put $k = n - z$, $\Lambda = az^p$, and

$$\begin{aligned} A &= ap(p-1)z^{p-2}G(\tau), & B &= \Lambda(\tau G'(\tau) + \tau^2 G''(\tau)), \\ C &= 1 + pk/z, & \Delta &= C^2 - AB, & \sigma^2 &= B/\Delta. \end{aligned} \quad (2.11)$$

For large n , $\Delta > 0$. Define

$$\mathcal{A}_{n,M} = \frac{R(\tau)}{\sqrt{\Delta}} \exp(\Lambda G(\tau) - k \log \tau). \quad (2.12)$$

We will prove, uniformly on the stated domain,

$$w_{n,M} \sim \mathcal{A}_{n,M}, \quad u_n(1) \sim \mathcal{A}_{n,M} e^{\nu_{n,M}}. \quad (2.13)$$

Thus the final formula needs a fixed finite implicit core, one two-variable saddle, and one increasing scalar equation. It never sums the original infinite kernel as an analytic function.

3 Exact formal algebra and probability

Lemma 3.1 (Positive coefficient identity). *For nonnegative slopes λ_j and amplitudes c_j with $c_1 = 1$, the formal equation $U = \sum_{j \geq 1} c_j q^j e^{\lambda_j U}$ has a unique solution with $[q]U = 1$. Its coefficients are*

$$[q^n]U = \sum_{m=1}^n \frac{1}{m!} \sum_{\substack{j_1 + \dots + j_m = n \\ j_i \geq 1}} \left(\prod_{i=1}^m c_{j_i} \right) \left(\sum_{i=1}^m \lambda_{j_i} \right)^{m-1}. \quad (3.1)$$

The zeroth power in the one-part term is interpreted as one.

Proof. The coefficient of degree n on the right uses coefficients of U only through degree $n - 1$, which proves triangular existence and uniqueness. Introduce an auxiliary marker v and solve $U_v = v \sum c_j q^j e^{\lambda_j U_v}$. Formal Lagrange inversion gives

$$[v^m]U_v = \frac{1}{m}[u^{m-1}] \left(\sum_{j \geq 1} c_j q^j e^{\lambda_j u} \right)^m.$$

Expanding the exponentials gives (3.1). Setting $v = 1$ is legitimate coefficientwise, because $m \leq n$ for the coefficient of q^n . $\square$

For the unmarked model $s = 1$, normalize the summands in (3.1) by $u_n(1)$. This defines a probability measure on ordered compositions of n . Write K_n for the number of parts, J_n for the largest part, and

$$E_n = n - K_n + 1 = 1 + \sum_i (j_i - 1), \quad N_{d,n} = \#\{i : j_i = d\}. \quad (3.2)$$

E_n is the total excess plus one; it is not generally the largest part. Marking in (2.2) gives the exact probability generating function

$$\mathbb{E}_s^{N_{d,n}} = \frac{u_n(s)}{u_n(1)}. \quad (3.3)$$

All distributions in this paper refer to this coefficient ensemble.

Let $\mathcal{E}_M$ be the event that exactly one part exceeds M .

Lemma 3.2 (Exact marking interpretation). *The branches in (2.4) are uniformly analytic near zero on compact core parameter sets, with nonnegative coefficients, and*

$$\mathbb{P}(\mathcal{E}_M) = w_{n,M}/u_n(1). \quad (3.4)$$

Proof. On a sufficiently small fixed polydisc, $\partial_u \Phi_M = O(q)$ uniformly. The analytic implicit function theorem, or a uniform contraction, gives G . Iteration from zero shows positivity. The geometric series defining R then has nonnegative coefficients as well.

Mark all actions above M by v . Differentiating the formal equation at $v = 0$ gives

$$(1 - \partial_u \Phi_M(q, G)) \partial_v U_v|_{v=0} = \sum_{j > M} q^j e^{a_j p G}.$$

The derivative is W_M . Its coefficient counts exactly one marked action in (3.1), proving (3.4). $\square$

4 Compact-uniform finite-core reduction

The following details justify using moving exponents $p = p_n$. They also isolate the role of positivity. No conclusion about a signed formal inverse is obtained by this argument.

4.1 Macroscopic localization

Let $L = \log n$ and

$$X_n = (p - 1)nL - pn \log L + \left(\log a + p \log \frac{p}{p - 1} - (p - 1) \right) n. \quad (4.1)$$

Lemma 4.1 (Uniform localization). *For the marked models with $s \in [0, S]$, $\log u_n(s) = X_n + o(n)$ uniformly. For any sufficiently small fixed $\varepsilon, \delta > 0$, the complement of*

$$|E_n L/n - p/(p-1)| \leq \delta, \quad J_n \geq (1-\varepsilon)E_n \quad (4.2)$$

has probability at most $\exp(-cn + o(n))$ for a uniform $c > 0$. The probability here is under the s -weighted coefficient measure, including $s = 0$. The same logarithmic equivalent holds for any fixed one-tail core.

Proof. Choose uniform constants D, W so that $\lambda_j \leq aj^p + D$ and $c_j \leq W$ for all j , while $c_1 = 1$. A composition with m parts and $r = n - m + 1$ has at most $r - 1$ nonunit parts. Convexity and successive transfers to a largest part give $\sum j_i^p \leq r^p + m - 1$. Its entire m -part row therefore has weight at most

$$\binom{n-1}{r-1} W^{r-1} \frac{(ar^p + D'm)^{m-1}}{m!}. \quad (4.3)$$

Conversely, a giant $r > d$ together with unit parts contributes at least $(ar^p)^{n-r}/(n-r)!$, independent of s .

Stirling's formula for the lower configuration at $r = \lfloor pn/((p-1)L) \rfloor$ gives $X_n + o(n)$, uniformly on compact parameter sets. We record the upper localization argument to avoid an unstated uniformity assumption. Rows with $r \geq \eta n$ lose a fixed multiple of nL for any fixed $\eta > 0$. Any remaining row within $O(n)$ of the lower bound must have $r/n \rightarrow 0$ and $\log r = L - O(\log L)$; otherwise its slope term is too small. Thus $D'm = o(r^p)$ and the binomial and amplitude entropies are $o(n)$. For $\alpha = r/n$, the preceding row estimate is $(1-\alpha)\{(p-1)L + p \log \alpha + \log a - \log(1-\alpha) + 1\} + o(1)$ after division by n . Comparing this with the lower bound first gives $\alpha L = O(\log L)$. Hence $\alpha \log \alpha = o(1)$ and the quadratic expansion errors also tend to zero. Writing $y = rL/n$, the row logarithm divided by n becomes

$$(p-1)L - p \log L + \log a + 1 + p \log y - (p-1)y + o(1). \quad (4.4)$$

The error is uniform in the localized range. Since $p \log y - (p-1)y$ tends uniformly to minus infinity at both ends of $(0, \infty)$, comparison with the lower bound now confines y to a fixed compact interval. This function has its unique maximum at $p/(p-1)$; its maximum gap outside a fixed neighborhood is bounded below uniformly for p in a compact interval. Summing at most n rows adds only $O(\log n)$.

If every part is at most $(1-\varepsilon)r$, convexity concentrates the excess into two extremal parts and bounds the slope sum by

$$ar^p((1-\varepsilon)^p + \varepsilon^p + o(1)).$$

The factor in parentheses is uniformly below one. Raising it to $m-1 \sim n$ gives an exponential loss that the $\exp(o(n))$ entropy cannot offset. This proves the giant estimate. The one-tail lower configuration and the full coefficient upper bound give the last assertion. $\square$

4.2 A uniform strict cutoff

Lemma 4.2 (Uniform sufficient core). *For a fixed core size L_0 containing all exceptional amplitudes and slopes, if $L_0(p_- - 1) > 1$, its one-tail coefficient divided by the full coefficient tends to one uniformly on the compact parameter set.*

Proof. On (4.2), the largest part J is unique. If a second part $i > L_0$ exists, replace (J, i) in their original slots by $(J+i-1, 1)$. The number of parts and their total sum stay fixed; the primitive amplitude product stays fixed because both large parts are in the unit-amplitude tail. For any fixed small $\eta > 0$, choosing ε sufficiently small gives the uniform slope gain

$$S' - S \geq a(p-\eta)r^{p-1}(i-1), \quad r = E_n.$$

Indeed, the mean-value theorem at the giant gives $ap(1 - \varepsilon)^{p-1}r^{p-1}(i - 1)$, and the lost slope of the smaller part is absorbed by making ε small. Also $S' \leq ar^p + D'm = ar^p(1 + o(1))$. Thus the old weight divided by the new one is at most

$$\exp\left(-\frac{(p - 2\eta)(n - r)(i - 1)}{r}\right).$$

Given an image, a chosen slot, and i , the preimage is recovered uniquely. There are at most n slots. Summing over $i > L_0$ bounds the bad weight by

$$n \sum_{i > L_0} \exp\left(-\frac{(p - 2\eta)(n - r)(i - 1)}{r}\right) \leq n^{1 - L_0(p-1) + \epsilon_0 + o(1)}$$

for an arbitrarily small uniform $\epsilon_0 > 0$, after narrowing the macroscopic window. The strict compact margin makes this tend to zero. The exceptional macroscopic set is exponentially negligible. The argument also works at zero values of exceptional amplitudes, because the transferred parts are beyond the exceptional core. $\square$

Apply this lemma to the core of size $d = M + 1$ in the marked model. Define

$$\begin{aligned} G_s(q) &= \Phi_M(q, G_s(q)) + sq^d e^{ad^p G_s(q)}, \\ R_s(q) &= \left(1 - \partial_u \Phi_M(q, G_s(q)) - sad^p q^d e^{ad^p G_s(q)}\right)^{-1}, \\ T_s(q) &= R_s(q) \sum_{j > d} q^j e^{aj^p G_s(q)}, \quad t_n(s) = [q^n] T_s(q). \end{aligned} \quad (4.5)$$

Then, uniformly for real $s \in [0, S]$,

$$u_n(s) = t_n(s)(1 + o(1)). \quad (4.6)$$

At $s = 0$, $G_s = G$ and $R_s = R$. The only difference between T_0 and W_M is the fixed $j = d$ summand. That summand is analytic near zero and has at most exponential coefficient growth. Since $\log w_{n,M} = X_n + o(n)$, it is negligible, and

$$t_n(0) \sim w_{n,M}. \quad (4.7)$$

5 Two saddles and their uniform perturbation

5.1 The shrinking Cauchy saddle

We provide the analytic estimates used later. They involve only a finite analytic core; the infinite tail is never exchanged with a contour integral.

For j in a compact multiple of $n/\log n$, put $k_j = n - j$ and $\Lambda_j = aj^p$. Let t_j solve

$$\Lambda_j t_j G'(t_j) = k_j. \quad (5.1)$$

Uniformly, $t_j \sim k_j/\Lambda_j \rightarrow 0$. Write $B_j = \Lambda_j(t_j G'(t_j) + t_j^2 G''(t_j)) \sim n$. Cauchy's formula gives

$$[q^{k_j}] R(q) e^{\Lambda_j G(q)} = \frac{R(t_j) e^{\Lambda_j G(t_j)} t_j^{-k_j}}{\sqrt{2\pi B_j}} (1 + o(1)). \quad (5.2)$$

Here and below these assertions hold also for G_s, R_s , uniformly in $s \in [0, S]$.

For completeness, on $q = t_j e^{i\theta}$ the linear phase vanishes by (5.1). The operator $D = q d/dq$ satisfies $\Lambda_j D^h G(t_j) = O(n)$ for each fixed positive h . On $|\theta| \leq n^{-2/5}$, Taylor expansion therefore

gives a Gaussian quadratic term and an $O(n^{-1/5})$ cubic error. Outside that arc, nonnegative Taylor coefficients and $[q]G = 1$ imply

$$\Re(G(t_j e^{i\theta}) - G(t_j)) \leq t_j(\cos \theta - 1).$$

Since $\Lambda_j t_j \asymp n$, the remaining integral is negligible. Uniform analyticity controls the amplitude R . This proves (5.2), including its compact-uniform version.

5.2 The giant-action saddle

Define

$$f_n(j) = \Lambda_j G(t_j) - (n - j) \log t_j.$$

Implicit differentiation of (5.1) gives

$$\begin{aligned} \frac{t'_j}{t_j} &= -\frac{C_j}{B_j}, \quad C_j = 1 + p(n - j)/j, \\ f'_n(j) &= apj^{p-1}G(t_j) + \log t_j, \\ f''_n(j) &= ap(p-1)j^{p-2}G(t_j) - C_j^2/B_j. \end{aligned} \quad (5.3)$$

Thus, uniformly for $j \asymp n/\log n$,

$$f''_n(j) = -\frac{pn}{j^2}(1 + o(1)), \quad f'''_n(j) = O\left(\frac{(\log n)^3}{n^2}\right). \quad (5.4)$$

At $j = cn/\log n$, $f'_n(j)/\log n = p/c - (p-1) + o(1)$. The endpoint signs and strict negative curvature give a unique saddle z in the coarse region, precisely (2.10). Its scales are

$$z \sim \frac{p}{p-1} \frac{n}{\log n}, \quad \tau \sim \frac{1}{a} \left(\frac{p-1}{p}\right)^p \frac{(\log n)^p}{n^{p-1}}. \quad (5.5)$$

The equivalences are uniform on compact parameter sets.

Macroscopic localization removes the complementary coarse region with exponentially small relative error. On $|j - z| \leq \sigma \log n$, with $\sigma^2 = B/\Delta \asymp n/(\log n)^2$, Taylor expansion gives

$$f_n(j) = f_n(z) - (j - z)^2/(2\sigma^2) + o(1).$$

The cubic error is $O((\log n)^3/\sqrt{n})$. Outside this window, curvature supplies an $\exp(-c(\log n)^2)$ loss. The prefactor in (5.2) varies by a relative $o(1)$ on the window. The lattice Gaussian sum is asymptotic to $\sqrt{2\pi}\sigma$. Consequently,

$$w_{n,M} = \mathcal{A}_{n,M}(1 + o(1)), \quad t_n(s) = \mathcal{A}_n(G_s, R_s)(1 + o(1)), \quad (5.6)$$

uniformly. The $\sqrt{2\pi}$ factors from the two saddles cancel. All assertions about uniqueness here concern the small-radius coarse region, not unrelated distant solutions of the saddle equations.

5.3 A normalized Hessian that remains invertible

The main new comparison can now be made at finite dimension. Define

$$H_G(j, y) = aj^p G(e^y) - (n - j)y. \quad (5.7)$$

The saddle equations say $\partial_j H_G = \partial_y H_G = 0$. Its Hessian at $(z, \log \tau)$ is

$$\begin{pmatrix} A & C \\ C & B \end{pmatrix}, \quad AB - C^2 = -\Delta.$$

It is indefinite, as expected from a Cauchy saddle followed by a maximum in the action variable. It must not be treated as a positive quadratic form.

Use local coordinates $\xi = (j - z)/z$ and $\eta = y - \log \tau$. The normalized Hessian is

$$\frac{1}{n} \begin{pmatrix} z^2 A & zC \\ zC & B \end{pmatrix} \longrightarrow \begin{pmatrix} p(p-1) & p \\ p & 1 \end{pmatrix}. \quad (5.8)$$

The limiting determinant is $-p$, uniformly bounded away from zero. Derivatives through any fixed order, in these normalized local coordinates, are $O(n)$ on a fixed small neighborhood. These facts permit ordinary quantitative implicit-function estimates even though the saddle is indefinite.

Lemma 5.1 (First omitted layer perturbs the optimized exponent linearly). *Let*

$$\mu = k\tau^M, \quad k = n - z, \quad (5.9)$$

using the M -core saddle. Uniformly for $s \in [0, S]$,

$$\log \frac{\mathcal{A}_n(G_s, R_s)}{\mathcal{A}_n(G, R)} = s\mu + O\left(n\tau^{M+1} + \frac{\mu^2}{n} + \frac{\mu}{n} + \tau^{M+1}\right). \quad (5.10)$$

Every error term tends to zero on (2.3).

Proof. Uniform analytic implicit differentiation, or a Taylor expansion of (4.5), yields

$$G_s(q) - G(q) = sq^d + O(q^{d+1}), \quad R_s(q) - R(q) = O(q^d), \quad d = M + 1. \quad (5.11)$$

The same orders hold after any fixed number of Euler derivatives. At the unperturbed saddle,

$$\begin{aligned} H_{G_s}(z, \log \tau) - H_G(z, \log \tau) &= s\Lambda\tau^d + O(\Lambda\tau^{d+1}) \\ &= sk\tau^{d-1} + O(n\tau^d), \end{aligned}$$

because $\Lambda\tau G'(\tau) = k$ and $G'(\tau) = 1 + O(\tau)$.

The gradient of this perturbation in (ξ, η) is $O(\mu)$. The inverse normalized Hessian is $O(1/n)$ by (5.8). The perturbed saddle therefore moves by

$$(z_s - z)/z = O(\mu/n), \quad \log(\tau_s/\tau) = O(\mu/n). \quad (5.12)$$

One can justify these bounds by applying the quantitative inverse-function argument on a ball of radius $C\mu/n$: the Hessian varies by $o(n)$ there, and the rescaled Newton map is a contraction. Uniqueness from (5.4) identifies the resulting saddle with the one already used in the coefficient formula. Evaluating at the displaced stationary point changes the optimized exponent by $O(\mu^2/n)$.

The relative changes of B , Δ , and the prefactor are $O(\mu/n + \tau^d)$, by the derivative bounds and (5.11). This proves (5.10).

Finally (5.5) gives, up to fixed logarithmic powers, $n\tau^d = O(n^{1-d(p_- - 1)}(\log n)^{dp_+}) \rightarrow 0$. Also $\mu^2/n \leq Cn\tau^{2M} \rightarrow 0$, since $2M(p_- - 1) > 1$: this follows from $d(p_- - 1) > 1$ and $2M \geq M + 1$. The remaining errors are smaller. All margins are strict and uniform. $\square$

5.4 Why the intensity loses all finite-core dependence

Lemma 5.2 (Additively accurate bare intensity). *With μ as in (5.9) and ν as in (2.7),*

$$\mu - \nu = O(nt_0^{M+1}) = o(1) \quad (5.13)$$

uniformly under (2.3).

Proof. For the bare core $G_0(q) = q$, the saddle equations are solved exactly by (j_0, t_0) in (2.7): $aj_0^p t_0 = k_0$ and $-\log t_0 = apj_0^{p-1} t_0 = pr$. Every allowed finite core satisfies $G(q) - q = O(q^2)$ uniformly. The perturbation of its optimized exponent is $O(nt_0)$, and its normalized gradient is $O(nt_0)$. Applying the same invertible-Hessian argument as above gives

$$(z - j_0)/j_0 = O(t_0), \quad \log(\tau/t_0) = O(t_0).$$

It follows that $k\tau^M - k_0 t_0^M = O(k_0 t_0^{M+1}) = O(nt_0^{M+1})$. The last expression tends to zero by the strict margin. $\square$

Remark 5.3 (Two different accuracies). It is not enough that $\mu/\nu \rightarrow 1$: both can diverge, and an $\exp(o(\nu))$ ambiguity would destroy a multiplicative coefficient formula. Lemma 5.2 gives the stronger *additive* $o(1)$. By contrast, the bare giant location j_0 need not be accurate on the Gaussian scale. The actual core saddle z remains the center in our joint limit theorem.

Proof of Theorem 2.1. Combine the uniform cutoff (4.6), the saddle formula (5.6), and Lemma 5.1:

$$u_n(s) = \mathcal{A}_{n,M} \exp(s\mu + o(1)).$$

The errors are uniform even at $s = 0$. By Lemma 5.2, replace μ by ν with additive $o(1)$ in the exponent. Finally $w_{n,M} \sim \mathcal{A}_{n,M}$. Exponentiating a uniform additive $o(1)$ proves the asserted relative estimate and also (2.13). $\square$

6 Every cutoff boundary has an explicit missing factor

Theorem 6.1 (Reciprocal-boundary repair). *Fix $M \geq 1$, $a > 0$, and set $p = 1 + 1/M$. Let $r_n > 0$ solve*

$$r_n^M (1 + r_n) e^{(M+1)r_n} = a^M n. \quad (6.1)$$

For any fixed allowed finite core,

$$u_n(1) \sim w_{n,M} \exp\left(\frac{r_n^{M+1}}{a^M}\right) \sim \mathcal{A}_{n,M} \exp\left(\frac{r_n^{M+1}}{a^M}\right). \quad (6.2)$$

Consequently

$$\log \frac{u_n(1)}{w_{n,M}} \sim \frac{(\log n)^{M+1}}{a^M (M+1)^{M+1}}. \quad (6.3)$$

Proof. Here $(M+1)(p-1) = (M+1)/M > 1$, so Theorem 2.1 applies. Raising (2.6) to the M th power gives (6.1). That equation gives the exact cancellation

$$\nu_{n,M} = \frac{nr_n}{1+r_n} e^{-(M+1)r_n} = \frac{r_n^{M+1}}{a^M}.$$

Substitution proves (6.2). Since $r_n \sim \log n/(M+1)$, the logarithmic equivalent follows. $\square$

Example 6.2 (The first two boundaries). For $M = 1$, $p = 2$, the extra factor beyond the one-action core is $\exp(r_n^2/a)$, where $r_n(1+r_n)e^{2r_n} = an$. This agrees with, and decomposes, the quadratic correction already found in [2]; it is a consistency check, not a new quadratic forward coefficient theorem.

For $M = 2$, $p = 3/2$, the two-action core misses $\exp(r_n^3/a^2)$, where $r_n^2(1+r_n)e^{3r_n} = a^2 n$. For $M = 3$, $p = 4/3$, the corresponding factor is $\exp(r_n^4/a^3)$. The same conclusion holds at every reciprocal boundary.

6.1 Why the leading logarithm is not a replacement for the root

Taking logarithms of (6.1), with $L = \log n$, gives

$$(M+1)r_n + M \log r_n + \log(1+r_n) = L + M \log a.$$

The first two scales are

$$r_n = \frac{L}{M+1} - \log L + \log(M+1) + \frac{M}{M+1} \log a + o(1). \quad (6.4)$$

Replacing r_n by $L/(M+1)$ changes r_n^{M+1}/a^M by order $L^M \log L$, which diverges. Even a relative asymptotic for r_n is therefore not a relative asymptotic for the missing factor. The scalar equation is a convenient exact resummation of all logarithmic terms needed for a coefficient equivalent.

7 The logarithmically shifted Poisson window

Theorem 7.1 (Critical window and finite survival fraction). *Fix $M \geq 1$, $a > 0$, and $\theta \in \mathbb{R}$. Let*

$$p_n = 1 + \frac{1}{M} + \frac{(M+1) \log \log n + \theta}{M \log n}. \quad (7.1)$$

Then

$$\nu_{n,M} \longrightarrow \lambda(\theta) := \frac{e^{-\theta}}{a^M (M+1)^{M+1}}, \quad \frac{w_{n,M}}{u_n(1)} \longrightarrow e^{-\lambda(\theta)}. \quad (7.2)$$

The number $N_{M+1,n}$ of first-omitted actions converges in total variation to $\text{Pois}(\lambda(\theta))$. The conclusions are uniform for θ in compact sets and for bounded allowed finite-core perturbations.

Proof. Let $p_0 = 1 + 1/M$. On compact parameter sets, $r_n \sim ((p-1)/p) \log n$. From (2.6) and (2.7),

$$\nu_{n,M} = a^{-M} n^{1-M(p-1)} r_n^{M+1} (1+r_n)^{M(p-1)-1}. \quad (7.3)$$

Equivalently,

$$\begin{aligned} \log \nu_{n,M} &= [1 - M(p-1)] \log n + Mp \log \log n - M \log a \\ &\quad + Mp \log \frac{p-1}{p} + o(1). \end{aligned} \quad (7.4)$$

The $o(1)$ is uniform for the particular moving window, as follows from the logarithmic root equation and $r_n/\log n \rightarrow 1/(M+1)$. Substitute (7.1). The $\log \log n$ terms cancel; terms involving $(p_n - p_0) \log \log n$ tend to zero. Thus

$$\log \nu_{n,M} \rightarrow -\theta - M \log a - (M+1) \log(M+1).$$

The coefficient fraction follows from Theorem 2.1; the probabilistic assertion is proved in Theorem 8.1 below. $\square$

Remark 7.2 (The window is not centered at the bare power threshold). At $p = p_0$ the intensity diverges like a positive power of $\log n$. Merely setting $p = p_0 + \theta/\log n$ still leaves that divergence. The center must first be moved upward by $((M+1)/M) \log \log n / \log n$. Only after this displacement does a width of order $1/\log n$ produce a finite transition profile.

The constant in (7.2) is easily absorbed into a centered coordinate. With

$$\theta = x - M \log a - (M+1) \log(M+1),$$

the limiting intensity is e^{-x} and the surviving one-tail fraction is $\exp(-e^{-x})$. This familiar profile is a consequence of a Poisson zero-count probability; no extremal-value interpretation is assumed.

7.1 Finite-size calibration

Formula (7.1) describes a limit, not a high-accuracy numerical calibration at moderate n . The term $(p_n - p_0) \log \log n$ in (7.4) can decay very slowly. For computation it is preferable to solve

$$\nu_{n,M}(a, p) = \lambda_* \quad (7.5)$$

using the scalar root (2.6), or to use the exact core intensity $\mu = k\tau^M$ if the core saddle is already available. This article does not assert global uniqueness of every finite- n calibration problem. The numerical routine brackets and checks the chosen local branch.

8 Poisson, Gaussian, and rare-event consequences

Theorem 2.1 and (3.3) give the strong real-marker law

$$\mathbb{E} s^{N_{d,n}} = \exp((s-1)\nu_{n,M} + o(1)), \quad 0 \leq s \leq S. \quad (8.1)$$

The remainder in the exponent is uniform. In particular it is an additive $o(1)$ even when the exponent itself diverges.

Theorem 8.1 (Distributional and rare-count laws). *Write $N_n = N_{M+1,n}$ and $\nu_n = \nu_{n,M}$ along any sequence allowed by (2.3).*

(i) *If ν_n stays bounded, then $d_{TV}(\mathcal{L}(N_n), \text{Pois}(\nu_n)) \rightarrow 0$.*

(ii) *If $\nu_n \rightarrow \infty$, then*

$$\frac{N_n - \nu_n}{\sqrt{\nu_n}} \implies \mathcal{N}(0, 1).$$

(iii) *For every regime,*

$$\mathbb{P}(N_n = 0) = \exp(-\nu_n)(1 + o(1)), \quad \mathbb{P}(\mathcal{E}_M \mid N_n = 0) \rightarrow 1.$$

(iv) *If $\nu_n \rightarrow \infty$, then for each fixed integer $m \geq 0$,*

$$\mathbb{P}(N_n = m) \sim e^{-\nu_n} \frac{\nu_n^m}{m!}.$$

Proof. For (i), first pass to any subsequence on which $\nu_n \rightarrow \lambda < \infty$. The pgfs converge on the positive real axis by (8.1). For any $S > 1$, positivity bounds their absolute values on $|s| \leq S$ by their values at S , which remain bounded. Every analytic subsequential limit therefore equals $\exp(\lambda(s-1))$ by the identity theorem. Cauchy's coefficient formula gives convergence of every point probability. The bound at $S > 1$ also gives uniform exponentially decreasing tails, so pointwise convergence upgrades to total variation. Compactness of the bounded sequence of intensities proves the claim without assuming that ν_n itself converges. This includes $\lambda = 0$.

For (ii), put $s = \exp(t/\sqrt{\nu_n})$ in (8.1) and subtract the centering term. For each bounded real t ,

$$\log \mathbb{E} \exp \left(t \frac{N_n - \nu_n}{\sqrt{\nu_n}} \right) = \nu_n \left(e^{t/\sqrt{\nu_n}} - 1 - \frac{t}{\sqrt{\nu_n}} \right) + o(1) \longrightarrow t^2/2.$$

The moment-generating-function continuity theorem proves the Gaussian limit.

Setting $s = 0$ proves the first part of (iii). For large n , configurations in $\mathcal{E}_M$ whose unique large part equals d have at most exponential total coefficient growth and are negligible relative to $w_{n,M}$. Apart from these, $\mathcal{E}_M$ implies $N_n = 0$. The identity $u_n(0) \sim w_{n,M}$ therefore proves the conditional claim, including when the conditioning event is exponentially rare in ν_n .

For (iv), define an analytic polynomial

$$F_n(t) = \frac{u_n(t/\nu_n)}{w_{n,M}}.$$

For every bounded positive real t , Theorem 2.1 gives $F_n(t) \rightarrow e^t$. Positivity gives $|F_n(t)| \leq F_n(R)$ on $|t| \leq R$. Normal-family compactness and uniqueness on the positive axis yield locally uniform analytic convergence to e^t . Therefore

$$[s^m]u_n(s) \sim w_{n,M} \frac{\nu_n^m}{m!}.$$

Divide by $u_n(1) \sim w_{n,M}e^{\nu_n}$. □

Remark 8.2 (What is not a consequence here). For diverging ν_n , we have not proved a total-variation approximation to $\text{Pois}(\nu_n)$ over *all* counts, a uniform local central limit theorem, or relative complex-marker control on fixed discs. The Gaussian limit and the fixed-count relative estimates do not by themselves imply any of those stronger statements.

Editorial note (ProveIt, 2026-09-29). For $M = 1$, $p = 2$, $a = 1$ the total-variation statement is proved in the microscopic-condensation package; see the editorial note in Section 11.

Theorem 8.3 (Poisson-rate large deviations). *If $\nu_n \rightarrow \infty$, the variables N_n/ν_n satisfy a large-deviation principle with speed ν_n and good rate function*

$$I(x) = \begin{cases} x \log x - x + 1, & x \geq 0, \\ +\infty, & x < 0, \end{cases} \quad I(0) = 1. \quad (8.2)$$

Proof. For every real t , (8.1) gives $\nu_n^{-1} \log \mathbb{E} e^{tN_n} \rightarrow e^t - 1$. Chernoff's inequality gives the upper bounds after optimizing over t ; for $x > 0$, the optimizing value is $t = \log x$ and yields (8.2). Positive t also gives exponential tightness.

For the matching lower bound at $x > 0$, tilt the measure by x^{N_n} . Its normalized moment generating function follows by taking the ratio of (8.1) at xe^t and at x . Chernoff bounds under this measure show $N_n/\nu_n \rightarrow x$ in probability. On a small interval around x , undoing the tilt costs $\exp(-\nu_n x \log x + o(\nu_n))$, while its normalizer is $\exp(\nu_n(x - 1) + o(\nu_n))$. Let the interval shrink to obtain the lower bound. At zero the exact relative law for $N_n = 0$ supplies the lower bound $-I(0) = -1$. Standard finite coverings on compact intervals, followed by exponential tightness, give the full closed-set/open-set formulation. □

9 The rare layer decouples from the giant action

Use the M -core saddle (z, τ) and $\sigma^2 = B/\Delta$ from (2.10)–(2.11). In particular,

$$\sigma^2 \sim \frac{z^2}{pn} \sim \frac{p}{(p-1)^2} \frac{n}{(\log n)^2}. \quad (9.1)$$

Theorem 9.1 (Uniform joint transform and independent limits). *For s in a compact subset of $[0, \infty)$ and real t in a bounded interval,*

$$\mathbb{E} \left[s^{N_{d,n}} \exp \left(t \frac{J_n - z}{\sigma} \right) \right] = \exp \left((s-1)\nu_{n,M} + \frac{t^2}{2} + o(1) \right) \quad (9.2)$$

uniformly on (2.3). Thus:

(i) If $\nu_n \rightarrow \lambda < \infty$, then

$$\left(N_{d,n}, \frac{J_n - z}{\sigma}\right) \Rightarrow (P, Z), \quad P \sim \text{Pois}(\lambda), \quad Z \sim \mathcal{N}(0, 1),$$

with independent limits.

(ii) If $\nu_n \rightarrow \infty$, then

$$\left(\frac{N_{d,n} - \nu_n}{\sqrt{\nu_n}}, \frac{J_n - z}{\sigma}\right) \Rightarrow (Z_1, Z_2),$$

where Z_1, Z_2 are independent standard Gaussians.

(iii) Conditional on $N_{d,n} = 0$, the standardized giant still converges to $\mathcal{N}(0, 1)$, even when $\mathbb{P}(N_{d,n} = 0) \rightarrow 0$.

Proof. First work in the one-tail representation for the augmented d -core. The largest action is its unique tail index j . Multiplying its outer saddle sum by $\exp(t(j - z)/\sigma)$ tilts a lattice Gaussian. Taylor expansion on the same window as before gives the factor

$$\exp\left(t \frac{z_s - z}{\sigma} + \frac{t^2 \sigma_s^2}{2\sigma^2}\right) (1 + o(1)).$$

By (5.12) and (9.1),

$$\frac{z_s - z}{\sigma} = O(\mu/\sqrt{n}) = o(1), \quad \sigma_s/\sigma \rightarrow 1.$$

Thus the factor is $\exp(t^2/2)(1 + o(1))$, uniformly in both markers.

We justify transferring this tilted estimate to the full positive coefficient model, rather than appealing to convergence in probability of the cutoff event alone. The giant tilt changes any weight by at most $\exp(O(n/\sigma)) = \exp(O(\sqrt{n} \log n))$, which cannot overcome the macroscopic exponential loss in Lemma 4.1. In the transfer map of Lemma 4.2, the giant changes by $i - 1$. The extra ratio is at most $\exp(|t|(i - 1)/\sigma)$. Its exponent per transferred unit is $O(\log n/\sqrt{n})$, negligible against the existing suppression of order $\log n$. The strict cutoff estimate therefore remains uniform for the positively tilted weights. This proves the required relative reduction.

Combining the tilted saddle factor with Theorem 2.1 proves (9.2). Substituting $s = e^v$ in the bounded-intensity case identifies the product limiting mgf. In the diverging case substitute $s = \exp(v/\sqrt{\nu_n})$ and center as in Theorem 8.1; the limit is $\exp((v^2 + t^2)/2)$. Finally set $s = 0$ in (9.2) and divide by the zero-count probability to obtain the conditional Gaussian mgf. $\square$

Remark 9.2 (The center must retain the core). The leading estimate $z \sim pn/((p - 1) \log n)$ does not determine a valid Gaussian center. Even ordinary next logarithmic corrections can exceed σ by an unbounded factor. Likewise the core-to-bare shift can be large relative to σ . Core independence of the scalar intensity ν does not imply core independence of the finely centered giant.

10 Computation and checks

10.1 Two independent exact coefficient algorithms

For the power slope $\lambda_j = aj^p$, set $E_{j,m} = [q^m] \exp(\lambda_j U(q))$. The triangular recurrence is

$$u_n = \sum_{j=1}^n c_j E_{j,n-j}, \quad E_{j,0} = 1, \quad E_{j,m} = \frac{\lambda_j}{m} \sum_{k=1}^m k u_k E_{j,m-k}. \quad (10.1)$$

For a finite core, compute G by the same recurrence with $c_j = 0$ above M . Then compute the coefficients of $P = \partial_u \Phi_M(q, G)$, $R = (1 - P)^{-1}$, and the convolution (2.5). All these calculations are finite to each requested order.

The independent check enumerates integer partitions. For a multiplicity vector (m_j) with $\sum j m_j = n$, its contribution is

$$\frac{\prod_j c_j^{m_j}}{\prod_j m_j!} \left(\sum_j \lambda_j m_j \right)^{\sum_j m_j - 1}. \quad (10.2)$$

To check the one-tail series, retain only vectors with $\sum_{j>M} m_j = 1$. This is not a recurrence enforcing the same identity; it is a separate finite enumeration based on the formal coefficient formula.

The supplied program uses exact rational arithmetic for $a = 1$, $p = 2, 3$, and degrees through 16. It compares the triangular recurrence with the partition sum for markers on action 2 or 3, at marker values 0, 1, 2; it separately compares the one-tail recurrence with the restricted partition sum for $M = 1, 2, 3$. All **288 exact assertions passed**. A further 32 comparisons of the log-domain numerical recurrence with the exact coefficients had maximum absolute logarithmic discrepancy 7.11×10^{-15} in the recorded run.

For example, at $a = 1, p = 2$, the first eight forward coefficients are

$$1, 2, \frac{15}{2}, \frac{107}{3}, \frac{1555}{8}, \frac{17362}{15}, \frac{5288311}{720}, \frac{15414767}{315}.$$

These checks detect normalization and marking mistakes. They do not prove an asymptotic theorem or certify a floating-point saddle.

10.2 Coefficient diagnostics with finite-size calibration

For each row of Table 1, the exponent p was chosen numerically so that the core intensity $\mu = k\tau^M$ equals $1/2$. The table compares exact-event coefficients computed by the positive log-domain recurrence, not by the asymptotic formula itself. Values are ordinary floating-point diagnostics, not outward-rounded enclosures.

Table 1: Calibrated coefficient diagnostics, $a = 1$. The prediction is $e^{-\nu}$ with the bare scalar intensity.

M	n	p	$w_{n,M}/u_n(1)$	$\mathbb{P}(N_{d,n} = 0)$	$e^{-\nu}$
1	80	2.562513	0.592485	0.595906	0.607610
1	160	2.554385	0.600168	0.601666	0.607037
1	320	2.539670	0.603573	0.604266	0.606770
2	80	1.707730	0.377341	0.449096	0.596771
2	160	1.728827	0.471136	0.508385	0.596605
2	320	1.739877	0.520370	0.541980	0.598192

The slower convergence for $M = 2$ is visible and should not be concealed. The normalized coefficient ratio $u_n/(\mathcal{A}_{n,M}e^\nu)$ is 1.02495, 1.01113, 1.00514 in the first three rows and 1.58321, 1.26638, 1.14950 in the last three. The theory asserts a limit, not that a minimal corrected core is already highly accurate at $n = 80$. The auxiliary CSV files retain the unrounded values and residuals.

10.3 A warning from the explicit window

The second diagnostic table evaluates the scalar/core saddles at very large n , using (7.1) with $\theta = -(M + 1)\log(M + 1)$ and $a = 1$. The limiting intensity is then one. Even at these large

values, the finite-size intensity need not be close to one: for $M = 1$, the computed core intensity is approximately 1.24932 at $n = 10^6$, 1.33102 at 10^{12} , and 1.23906 at 10^{48} . This is consistent with slowly vanishing logarithmic corrections; it is not evidence of a monotone convergence theorem. The full diagnostic file records $M = 1, 2, 3$. For numerical work the root-defined ν or the calibrated μ should be used instead of the limiting constant alone.

10.4 Reproducibility and complexity

The bundle contains `verification/verify.py`, recorded JSON/CSV outputs, and a run log. The exact checks require Python's standard rational arithmetic; the same script also imports NumPy and SciPy for the numerical part. The recorded environment used Python 3.13.5, NumPy 2.3.5, and SciPy 1.17.0. The direct triangle takes $O(N^3)$ arithmetic operations and $O(N^2)$ storage; this is not an exact-arithmetic bit-complexity estimate.

The saddle routine checks the implicit residual, the regular core branch, and the determinant sign. These checks are necessary but not interval certificates. A small floating residual alone is not a proof that a numerical root has been enclosed. No external service is used by the verification program, and no repository file was modified in producing this package.

11 Further research questions

11.1 A full local Poisson approximation at diverging intensity

The real-marker theorem gives a central limit theorem, fixed rare-count equivalents, and large deviations. Prove a uniform local approximation for $m = \nu + O(\sqrt{\nu} \log \nu)$ and then total-variation convergence to $\text{Pois}(\nu)$ when $\nu \rightarrow \infty$. A promising approach is to extract the marker coefficient within the two-saddle calculation, controlling the conditioning correction ν^2/n directly. This would be stronger than merely invoking the Gaussian weak limit.

Editorial note (ProveIt, 2026-09-29). Answered for $M = 1, p = 2, a = 1$, that is $\lambda_j = j^2$ with any nonnegative first slope, by the independent package `Microscopic_Condensation_Exponential_Feedback/` (its Theorem `thm:probability` and Corollary `cor:local`): the number of twos, whose intensity $r_n^2 = \nu_{n,1}$ diverges, is close in total variation to $\text{Pois}(r_n^2)$ jointly with, and asymptotically independently of, a discrete Gaussian largest action, with a two-dimensional local limit. Other (M, p, a) remain open.

11.2 The next omitted layer and interaction corrections

Our compact margin makes $n\tau^{M+1}$ and μ^2/n vanish. At the next boundary these terms can survive. Determine the joint law of actions $M + 1$ and $M + 2$ and the first interaction correction to the exponential factor. The normalized Hessian in (5.8) gives a concrete starting point: the optimized second-order correction is a quadratic form in the marked perturbation gradients. It need not have the sign of a covariance matrix, because the underlying two-variable saddle is indefinite.

11.3 Fixed logarithmic corrections to the power tail

For slopes $aj^p(\log j)^\gamma$, the repository predicts a new cutoff behavior at $M(p - 1) = 1$. Extend the compact-uniform transfer and saddle estimates to a differentially regularly varying tail. The delicate target is the finite correction at $\gamma = p$, including its constant and Poisson parameter. The present moving- p theorem does not prove this fixed-tail statement, although its perturbation mechanism suggests what to measure.

11.4 All finite cloud coordinates and the correct centers

Study the joint vector of counts of actions $2, \dots, M + 1$, together with the giant. Some counts diverge on different scales, and leading equivalents for their means may be too inaccurate for central limits. Introduce several markers in the finite core, solve the corresponding marked saddle equations, and compute the complete covariance and centering corrections. The present paper resolves the first omitted coordinate, not this whole multiscale vector.

11.5 The signed quadratic inverse conjecture

The source [2] conjectures a sharp equivalent for the formal inverse at $p = 2$. Its exact inverse one-tail response is known, but positivity is lost. A signed finite-core reduction must control cancellations before a Poisson-type exponent can be trusted. Our transfer estimates cannot simply be applied with absolute values: doing so may swamp the proposed inverse coefficient. An analytic-core regrouping of the inverse Lagrange formula is still a focused and worthwhile next problem.

Editorial note (ProveIt, 2026-09-29). Two later packages claim independent proofs of this conjecture, neither yet reviewed: `Quadratic_Exponential_Feedback_After_Reversion/` and `Signed_Quadratic_Feedback_Inversion/` (Theorem `thm:main` in each). The first identifies its relative error with this article's layer intensity divided by r_n .

11.6 Quantified error bounds and certified coefficient budgets

Replace the compact-uniform $o(1)$ terms by explicit bounds with numerical onset values. The perturbation estimates already identify two obstruction scales, $n\tau^{M+1}$ and ν^2/n ; the cutoff and contour bounds must supply computable constants as well. Combine these with interval evaluation of the finite core and a certified saddle enclosure. The objective is an actual coefficient interval, not merely a residual-based diagnostic.

11.7 A growing core near the linear-feedback boundary

As $p \downarrow 1$, the required core size grows. A theorem allowing $M = M_n \rightarrow \infty$ would require uniform analytic radii and uniform control of all finite-core derivatives. The present proof fixes M throughout. The interaction between a growing core and the scale $(p - 1) \log n$ is not captured by taking a formal limit in (2.8).

11.8 Formalization and analytic realization

A useful Lean development can begin with the finite marking identities, the scalar monotonicity in (2.6), the exact boundary cancellation, and a reusable parameter-uniform saddle-perturbation lemma. The later probability and contour arguments should be separate modules with explicit hypotheses. In parallel, investigate whether these coefficient-layer laws control optimally truncated analytic realizations on the negative ray. A least formal term or a coefficient probability law is not automatically an analytic remainder theorem.

12 Conclusion

The strict finite-core cutoff leaves a quantitatively structured remainder. In the range where one more core action would suffice, the entire first omitted layer contributes a single exponential

factor with an explicit scalar intensity. That intensity is insensitive, to additive $o(1)$, to bounded nonnegative changes inside the core.

At every reciprocal boundary the correction grows like the exponential of a power of $\log n$. A finite nonzero survival fraction emerges only after a $\log \log n / \log n$ displacement of the exponent. The same coefficient theorem yields Poisson limits, Gaussian fluctuations, exponentially small fixed-count probabilities, and a joint limit in which the rare layer and the giant fluctuation decouple.

These results extend specific repository questions without confusing formal coefficient analysis with analytic summability, signed reversion, or a machine-checked development. The unresolved extensions above are stated as separate mathematical tasks rather than conclusions inferred by analogy.

A Proof-dependency and scope audit

Statement	Main inputs	Status here
Positive coefficients and one-tail identity	Formal Lagrange inversion; finite marking	Proved; classical tools
Compact-uniform sufficient core	Positive transfer, preimage count, uniform convex localization	Re-established from repository method
Uniform finite-core saddle	Cauchy Gaussian estimate, action saddle, lattice summation	Re-established in needed scope
Marked exponential repair	Uniform core jet and invertible normalized indefinite Hessian	Main extension
Core-independent intensity	Additive bare-saddle comparison	Main extension
Boundary correction and shifted window	Exact scalar cancellations and uniform theorem	Main extension
Bounded-intensity total variation	Positive pgfs, analytic compactness, exponential tightness	Proved
Diverging-intensity CLT and fixed counts	Real-marker theorem; scaled analytic compactness	Proved
Giant/layer independence	Positive tilt, uniform cutoff, shifted Gaussian saddle	Proved
Signed inverse; general regular variation; growing core	Additional cancellation or uniformity arguments required	Not claimed

The full power tail is exact outside the specified finite core. A merely asymptotically equivalent tail, unbounded signed amplitudes, a growing M , or approach to the excluded lower boundary require additional work. All asymptotic assertions are as $n \rightarrow \infty$. Computations in this package are checks of formulas and diagnostics, not additional hypotheses in a proof.

B Notation and an implementation checklist

Symbol	Meaning
$M, d = M + 1$	Core size and first omitted primitive action
$\widehat{U}_s, u_n(s)$	Formal marked feedback solution and its n th coefficient
G, R	Analytic finite-core solution and linear response
$W_M, w_{n,M}$	Formal exactly-one-tail series and coefficient
G_s, R_s, T_s	Augmented d -core and its exactly-one-tail response
z, τ, k	Actual M -core saddle action, radius, and $n - z$
r, j_0, t_0, k_0	Scalar bare root and bare saddle quantities
$\mu = k\tau^M$	Core-dependent intensity used in the perturbation proof
$\nu = k_0 t_0^M$	Explicit core-independent intensity; $\mu - \nu = o(1)$
A, B, C, Δ	Hessian entries and positive saddle determinant factor
$\mathcal{A}_{n,M}$	Finite-core saddle equivalent for $w_{n,M}$
$N_{d,n}$	Number of primitive action- d occurrences in the coefficient ensemble
J_n, σ	Largest action and its Gaussian scale

For numerical use, first choose a core with $(M + 1)(p - 1) > 1$ and a positive margin. Solve (2.6) in logarithms, solve the finite-core saddle on its regular small branch, and evaluate (2.12) in logarithms. Add ν to its logarithm, not a leading asymptotic replacement for ν . Check the branch and residual, but do not interpret those checks as an interval enclosure. Increase the core or use a more accurate correction when the finite-size residual scales are not small.

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